

Computer Networks

UNIT I

Data communications refers to the transmission of digital data between two or more computers. The physical connection between networked computing devices is established using either cable media or wireless media. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). The effectiveness of a data communications system depends on four fundamental characteristics: **delivery, accuracy, timeliness, and jitter.**

Delivery: The system must deliver data to the correct destination. Data must be received by the intended device or user

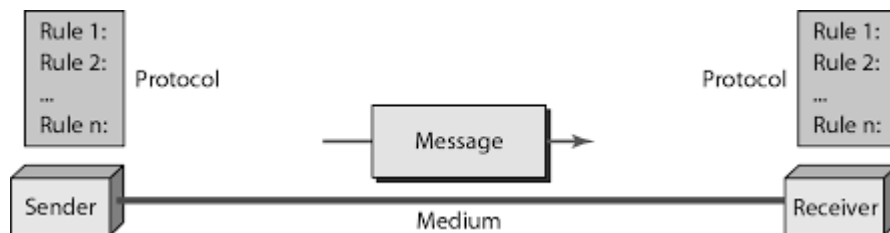
Accuracy: The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected or unusable.

Timeliness: The system must deliver data in a timely manner. Data that have been altered are useless

Jitter: Jitter refers to the variation in packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Components:

1. A data communications system has five components.



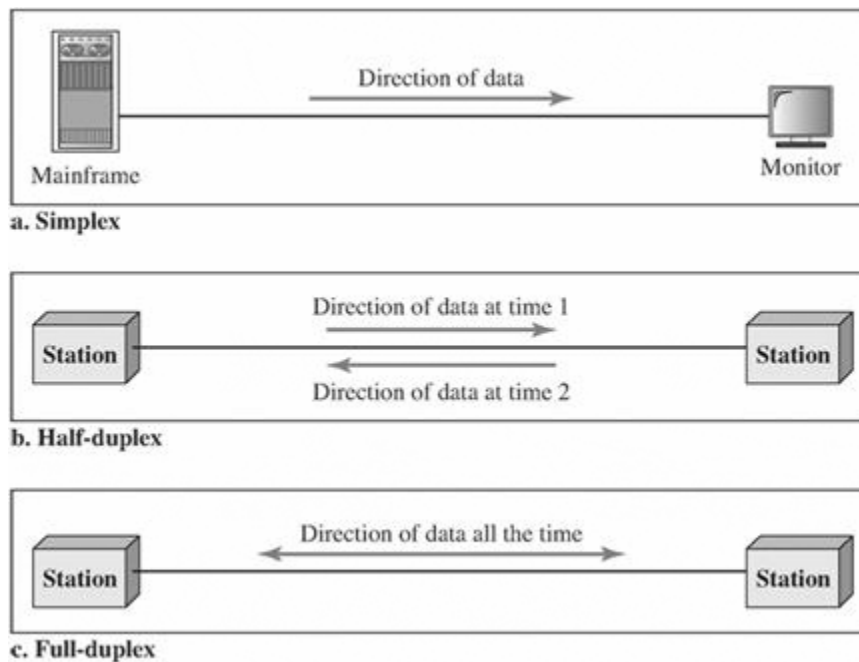
Introduction to Data Communications:

A data communications system has five components:

1. **Message:** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender:** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver:** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves
5. **Protocol:** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Direction of Data Flow



Simplex: In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction.

Half-Duplex: In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is like a one-lane road with traffic allowed in both directions.

Full-Duplex: In full-duplex both stations can transmit and receive simultaneously. The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction.

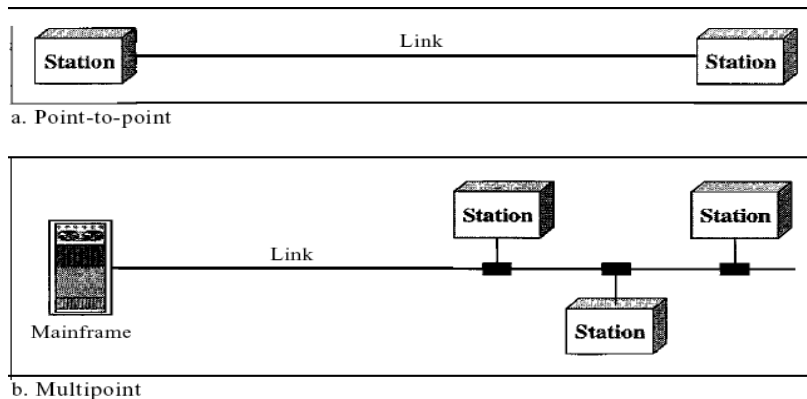
Type of Connection A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.

For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections: point-to-point and multipoint.

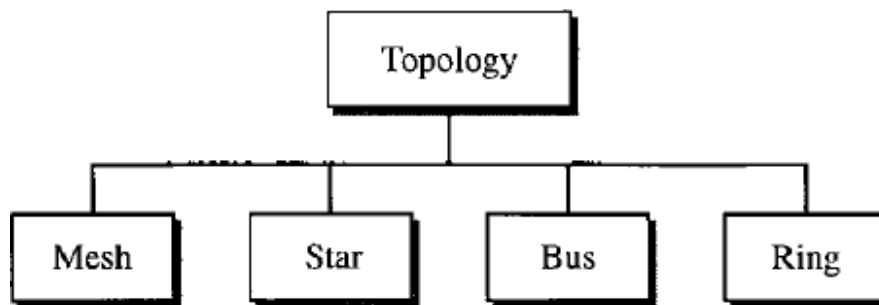
Point-to-Point A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point

connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible

Multipoint A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection.



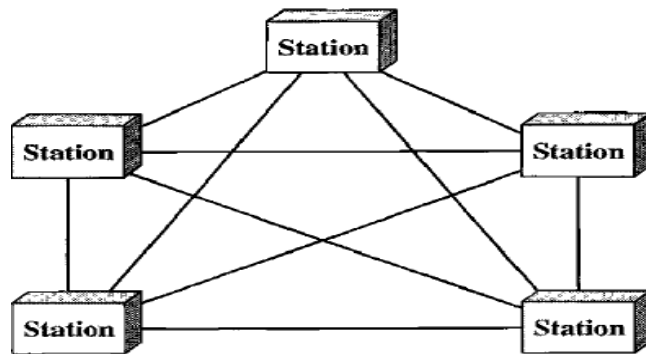
Physical Topology The term physical topology refers to the way in which a network is laid out physically. One or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus, and ring



Mesh: In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node. Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need $n(n - 1) / 2$ duplex-mode links.

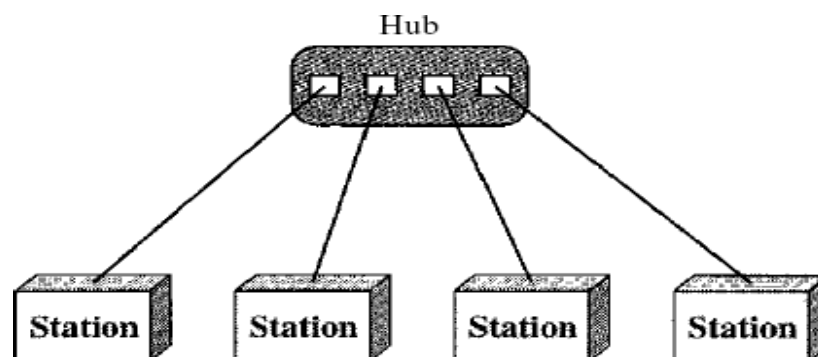
Advantages: 1. The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.

2. A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system. Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.



Star Topology: In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices. The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

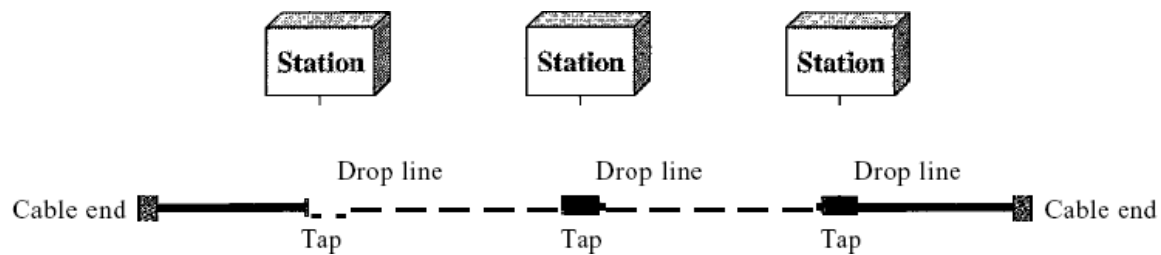
One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead. Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).



Bus Topology: The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.

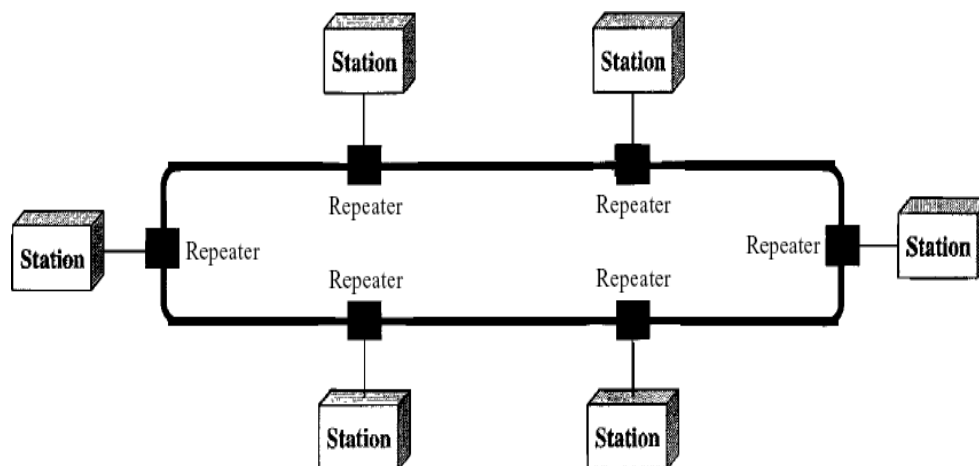
Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and

on the distance between those taps.



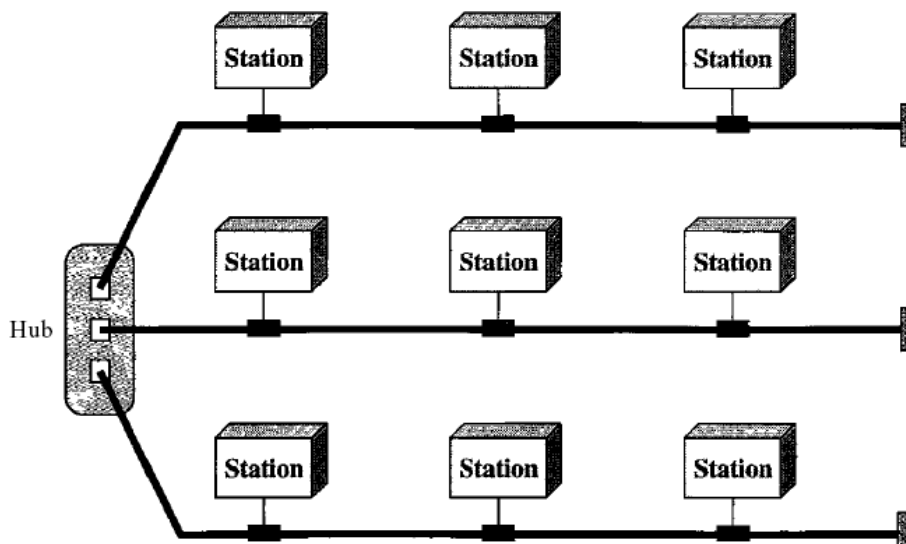
Advantages of a bus topology include ease of installation. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

Ring Topology In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.



A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations (maximum ring length and number of devices).

Hybrid Topology A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology as shown in Figure



Categories of Network

Local area networks, generally called LANs, are privately-owned networks within a single building or campus of up to a few kilometres in size. They are widely used to connect personal computers and workstations in company offices and factories to share resources (e.g., printers) and exchange information. LANs are distinguished from other kinds of networks by three characteristics:

- (1) Their size,
- (2) Their transmission technology, and
- (3) Their topology.

LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance. Knowing this bound makes it possible to use certain kinds of designs that would not otherwise be possible. It also simplifies network management. LANs may use a transmission technology consisting of a cable to which all the machines are attached, like the telephone company party lines once used in rural areas. Traditional LANs run at speeds of 10 Mbps to 100 Mbps, have low delay (microseconds or nanoseconds), and make very few errors. Newer LANs operate at up to 10 Gbps. Various topologies are possible for broadcast LANs.

Metropolitan Area Network:

A metropolitan area network, or MAN, covers a city. The best-known example of a MAN is the cable television network available in many cities. This system grew from earlier community antenna systems used in areas with poor over-the-air television reception. In these early systems, a large antenna was placed on top of a nearby hill and signal was then piped to the subscribers' houses. At first, these were locally-designed, ad hoc systems. Then companies began jumping into the business, getting contracts from city governments to wire up an entire city.

Wide Area Network:

A wide area network, or WAN, spans a large geographical area, often a country or continent. It

contains a collection of machines intended for running user (i.e., application) programs. These machines are called as hosts. The hosts are connected by a communication subnet, or just subnet for short. The hosts are owned by the customers (e.g., people's personal computers), whereas the communication subnet is typically owned and operated by a telephone company or Internet service provider. The job of the subnet is to carry messages from host to host, just as the telephone system carries words from speaker to listener.

Protocols and Standards

In computer networks, communication occurs between entities in different systems. An entity is anything capable of sending or receiving information. However, two entities cannot simply send bit streams to each other and expect to be understood. For communication to occur, the entities must agree on a protocol. **A protocol is a set of rules** that govern data communications. A protocol defines what is communicated, **how** it is communicated, and **when** it is communicated. The key elements of a protocol are syntax, semantics, and timing.

- o **Syntax.** The term *syntax* refers to the structure or format of the data, meaning the order in which they are presented. For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself.

- o **Semantics.** The word *semantics* refers to the meaning of each section of bits. How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation? For example, does an address identify the route to be taken or the final destination of the message?

- o **Timing.** The term *timing* refers to two characteristics: when data should be sent and how fast they can be sent. For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

Standards

Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunications technology and processes. Standards provide guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.

Data communication standards fall into two categories: *de facto* (meaning "by fact" or "by convention") and *de jure* (meaning "by law" or "by regulation").

o De facto. Standards that have not been approved by an organized body but have been adopted as standards through widespread use are de facto standards. De facto standards are often established originally by manufacturers who seek to define the functionality of a new product or technology.

o De jure. Those standards that have been legislated by an officially recognized body are de jure standards.

Networking hardware

Networking hardware, also known as **network equipment** or **computer networking devices**, are electronic devices which are required for communication and interaction between devices on a [computer network](#).

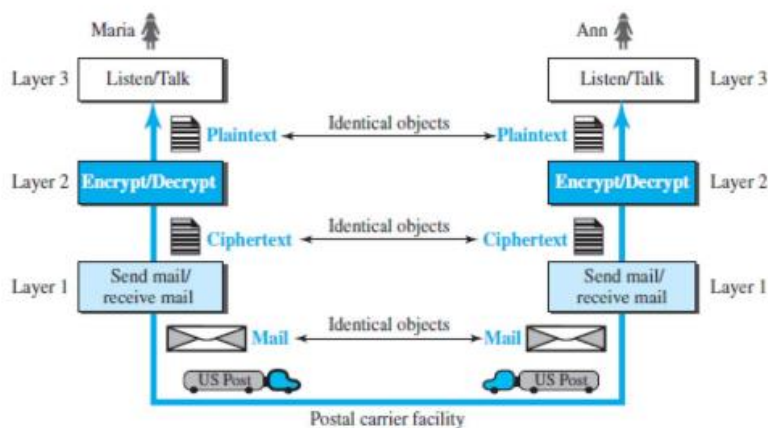
Network software

Network software encompasses a broad range of software used for design, implementation, and operation and monitoring of computer networks. Traditional networks were hardware based with software embedded. With the advent of Software Defined Networking (SDN), software is separated from the hardware thus making it more adaptable to the ever-changing nature of the computer network.

Functions of Network Software

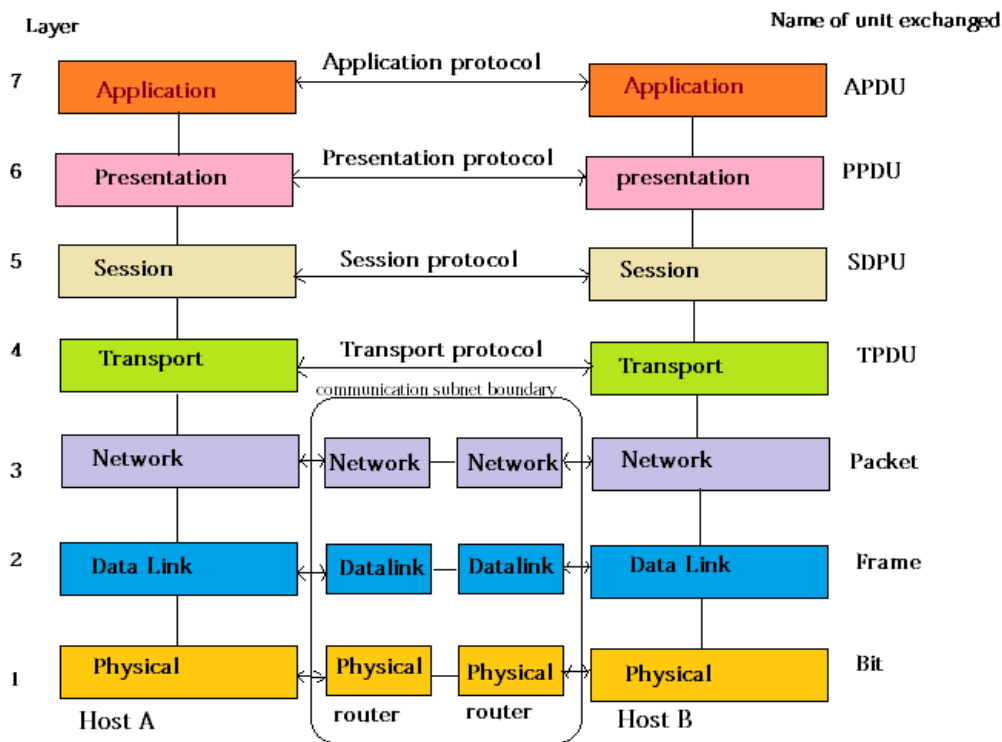
- Helps to set up and install computer networks
- Enables users to have access to network resources in a seamless manner
- Allows administrations to add or remove users from the network
- Helps to define locations of data storage and allows users to access that data
- Helps administrators and security system to protect the network from data breaches, unauthorized access and attacks on a network
- Enables network virtualizations

Layering Scenario



THE OSI MODEL

Open Systems Interconnection Basic Reference Model (OSI Reference Model or OSI Model) is an abstract description for layered communications and computer network protocol design. It was developed as part of the Open Systems Interconnection (OSI) initiative. In its most basic form, it divides network architecture into seven layers which, from top to bottom, are the Application, Presentation, Session, Transport, Network, Data-Link, and Physical Layers. It is therefore often referred to as the OSI



Seven Layer Model.

1. Physical layer

- One of the major function of the physical layer is to move data in the form of electromagnetic signals across a transmission medium.
- Its responsible for movements of individual bits from one hop (Node) to next.
- Both data and the signals can be either *analog* or *digital*.
- 4. Transmission media work by conducting energy along a physical path which can be wired or wireless

Concerned:

- **Physical characteristics of interface and medium** :The physical layer defines the characteristics of the interface between the devices and the transmission medium.
- **Representation of bits**: The physical layer data consists of a stream of bits (sequence of 0s or 1s) with no interpretation. To be transmitted, bits must be encoded into signals—electrical.
- **Data rate**: The transmission rate-the number of bits sent each second-is also defined by the physical layer
- **Synchronization of bits**: The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level
- **Line configuration**: The physical layer is concerned with the connection of devices to the media.

In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.

- Physical topology: The physical topology defines how devices are connected to make a network.
- Transmission mode: The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

2.Data link layer

This is responsible for moving frames from one hop (Node) to the next.

Concerned:

- Framing: The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- Physical addressing: If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame
- Flow Control: If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism.
- Error Control: The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames.
- Access Control: When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time

3. Network layer

This layer is responsible for the delivery of individual packets from the source host to the destination host.

Concerned:

- Logical addressing: The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver
- Routing When independent networks or links are connected to create internetworks (network of networks) or a large network, the connecting devices (called routers or switches) route or switch the packets to their final destination.

4. Transport layer

The transport layer is responsible for the delivery of a message from one process to another

Concerned:

- Service-point addressing The transport layer header must include a type of address called a service-point address (or port address). The network layer gets each packet to the correct computer.
- Segmentation and reassembly: A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.
- Connection control The transport layer can be either connectionless or connection oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection oriented transport layer makes a connection with the transport layer at the destination

- Flow control: the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link
- Error Control: Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to-process rather than across a single link.

5. Session layer

The session layer is responsible for dialog control and synchronization

Concerned:

- Dialog Control: The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either halfduplex (one way at a time) or full-duplex (two ways at a time) mode.
- Synchronization: The session layer allows a process to add checkpoints, or synchronization points, to a stream of data.

6. Presentation layer

The presentation layer is responsible for translation, compression and encryption

Concerned:

- Translation The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on.
- Encryption To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network.
- Compression: Data compression reduces the number of bits contained in the information

7. Application layer

The application layer is responsible for providing services to the user.

Concerned:

- Network virtual terminal. A network virtual terminal is a software version of a physical terminal, and it allows a user to log on to a remote host. File transfer, access and management
- File transfer, access, and management. This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local computer, and to manage or control files in a remote computer locally.
- Mail services. This application provides the basis for e-mail forwarding and storage.
- Directory services. This application provides distributed database sources and access for global information about various objects and services.

TCP/IP PROTOCOL SUITE

The Transmission Control Protocol/ Internet Protocol suite of protocols form the basis of the internet. It is TCP/IP that creates a virtual network when multiple computer networks are connected together. The TCP/IP network was earlier known as ARPANET, but is now known as internet.

TCP/IP Suite consists of Four layer

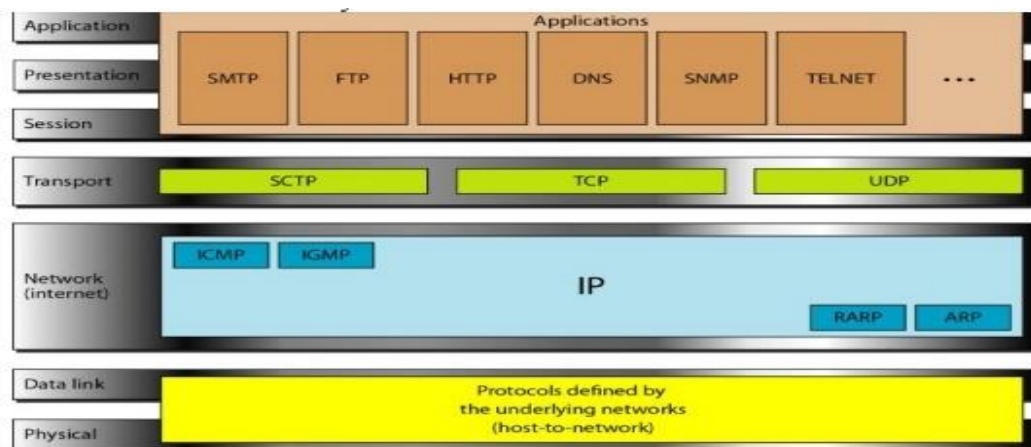
Network Interface: - It include the function of physical layer and data link layer. TCP/IP protocol suite includes Host-to-network layer protocols such as Serial Line internet protocol and point to point protocol.

Internet Layer: - The internet layer is exactly same to the network layer of OSI model. IP is the primary protocol operating at this layer and it provides data encapsulation routing, addressing and fragmentation services to the protocols at the transport layer above it.

Transport Layer: - TCP/IP Suite includes two protocol at this layer, the transmission control protocol and the user datagram protocol These protocol provides connection or connectionless data transfer services.

Application layer: - The TCP/IP protocol at the application layer can take different forms of Protocols, such as the File Transfer Protocol, Hypertext Transfer Protocol.

TCPI/IP and OSI model



Application Layer The application layer in TCPIIP is equivalent to the combined session, presentation, and application layers in the OSI model

SMTP. Stands for "Simple Mail Transfer Protocol." This is the protocol used for sending e-mail over the Internet.

FTP File Transfer Protocol (FTP) is a standard Internet protocol for transmitting files between computers on the Internet over TCP/IP connections

The Hypertext Transfer Protocol (**HTTP**) is an application-level protocol for distributed, collaborative, hypermedia information systems.

The Domain Name System (**DNS**) is a hierarchical and decentralized naming system for **computers**, services, or other resources connected to the Internet or a private **network**.

Simple Network Management Protocol (SNMP) is an Internet Standard protocol for collecting and organizing information about managed devices on IP networks.

Telnet is an application protocol used on the Internet or local area **network** to provide a bidirectional interactive text-oriented communication facility using a virtual terminal connection

Transport Layer

Traditionally the transport layer was represented in TCP/IP by two protocols: TCP and UDP. IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another.

Stream Control Transmission Protocol The Stream Control Transmission Protocol (SCTP) provides support for newer applications such as voice over the Internet. It is a transport layer protocol that combines the best features of UDP and TCP.

The Transmission Control Protocol (TCP) provides full transport-layer services to applications. TCP is a reliable stream transport protocol. The term stream, in this context, means connection-oriented.

User Datagram Protocol The User Datagram Protocol (UDP) is the simpler of the two standard TCPIIP transport protocols. It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

Network Layer

At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

Internet Control Message Protocol The Internet Control Message Protocol (ICMP) is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender. ICMP sends query and error reporting messages

Internet Group Message Protocol The Internet Group Message Protocol (IGMP) is used to facilitate the simultaneous transmission of a message to a group of recipients.

Internetworking Protocol (IP) The Internetworking Protocol (IP) is the transmission mechanism used by the TCP/IP protocols. It is an unreliable and connectionless protocol-a best-effort delivery service.

Reverse Address Resolution Protocol The Reverse Address Resolution Protocol (RARP) allows a host to discover its Internet address when it knows only its physical address. It is used when a computer is connected to a network for the first time or when a diskless computer is booted.

Address Resolution Protocol The Address Resolution Protocol (ARP) is used to associate a logical address with a physical address. On a typical physical network, such as a LAN, each device on a link is identified by a physical or station address, usually imprinted on the network interface card (NIC).

Physical and Data Link Layers

At the physical and data link layers, TCPIIP does not define any specific protocol. It supports all the standard and proprietary protocols. A network in a TCPIIP internetwork can be a local-area network or a wide-area network.

INTERNET

The Internet has revolutionized many aspects of our daily lives. It has affected the way we do

business as well as the way we spend our leisure time. Count the ways you've used the Internet recently. Perhaps you've sent electronic mail (e-mail) to a business associate, paid a utility bill, read a newspaper from a distant city, or looked up a local movie schedule-all by using the Internet. Or maybe you researched a medical topic, booked a hotel reservation, chatted with a fellow Trekkie, or comparison-shopped for a car. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

A Brief History

A network is a group of connected communicating devices such as computers and printers. An internet (note the lowercase letter i) is two or more networks that can communicate with each other. The most notable internet is called the Internet (uppercase letter I), a collaboration of more than hundreds of thousands of interconnected networks. Private individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet.

In the mid-1960s, mainframe computers in research organizations were standalone devices. Computers from different manufacturers were unable to communicate with one another. The **Advanced Research Projects Agency (ARPA)** in the Department of Defense (DoD) was interested in finding a way to connect computers so that the researchers they funded could share their findings, thereby reducing costs and eliminating duplication of effort.

In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for ARPANET, a small network of connected computers. The idea was that each host computer (not necessarily from the same manufacturer) would be attached to a specialized computer, called an *interface message processor* (IMP). The IMPs, in turn, would be connected to one another.

In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the *Internetworking Project*. Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of packets. This paper on Transmission Control Protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway. Shortly thereafter, authorities made a decision to split TCP into two protocols: Transmission Control Protocol (TCP) and Internetworking Protocol (IP).

The Internet Today

The Internet has come a long way since the 1960s. The Internet today is not a simple hierarchical structure. It is made up of many wide- and local-area networks joined by connecting devices and switching stations.

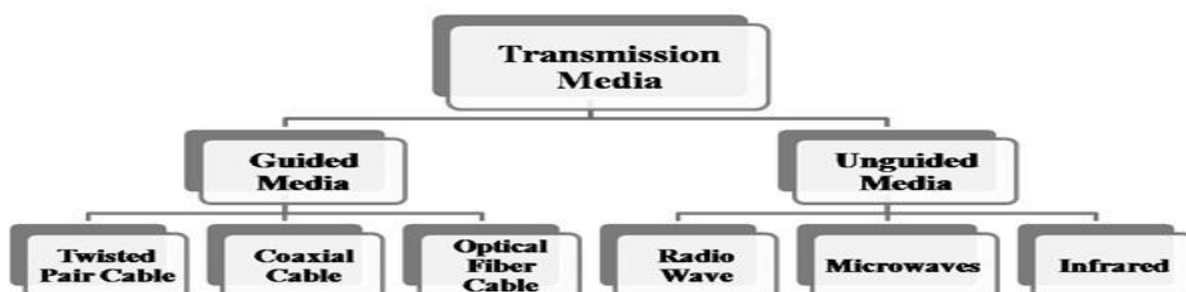
Today most end users who want Internet connection use the services of Internet service providers (ISPs). There are international service providers, national service providers, regional service providers and local service providers. The Internet today is run by private companies, not the government.

Difference between TCP/IP and OSI Model

Following are the differences between OSI and TCP/IP Reference Model –

OSI	TCP/IP
OSI represents Open System Interconnection .	TCP/IP model represents the Transmission Control Protocol / Internet Protocol.
OSI is a generic, protocol independent standard. It is acting as an interaction gateway between the network and the final-user.	TCP/IP model depends on standard protocols about which the computer network has created. It is a connection protocol that assigns the network of hosts over the internet.
The OSI model was developed first, and then protocols were created to fit the network architecture's needs.	The protocols were created first and then built the TCP/IP model.
It provides quality services.	It does not provide quality services.
The OSI model represents defines administration, interfaces and conventions. It describes clearly which layer provides services.	It does not mention the services, interfaces, and protocols.
The protocols of the OSI model are better unseen and can be returned with another appropriate protocol quickly.	The TCP/IP model protocols are not hidden, and we cannot fit a new protocol stack in it.
It is difficult as distinguished to TCP/IP.	It is simpler than OSI.
It provides both connection and connectionless oriented transmission in the network layer; however, only connection-oriented transmission in the transport layer.	It provides connectionless transmission in the network layer and supports connecting and connectionless-oriented transmission in the transport layer.
It uses a horizontal approach.	It uses a vertical approach.
The smallest size of the OSI header is 5 bytes.	The smallest size of the TCP/IP header is 20 bytes.
Protocols are unknown in the OSI model and are returned while the technology modifies.	In TCP/IP, returning protocol is not difficult.

Transmission Media

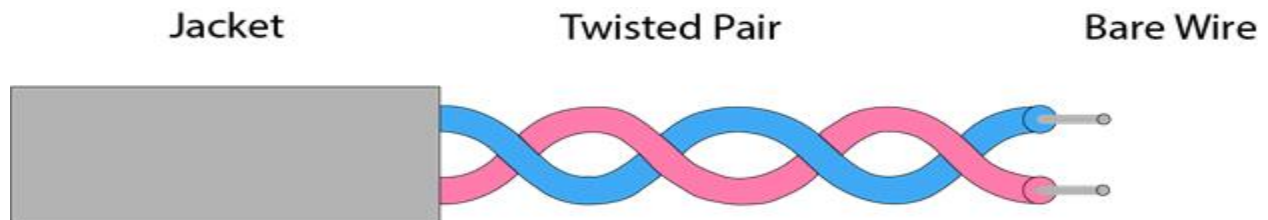


Guided Media It is defined as the physical medium through which the signals are transmitted. It is also known as Bounded media.

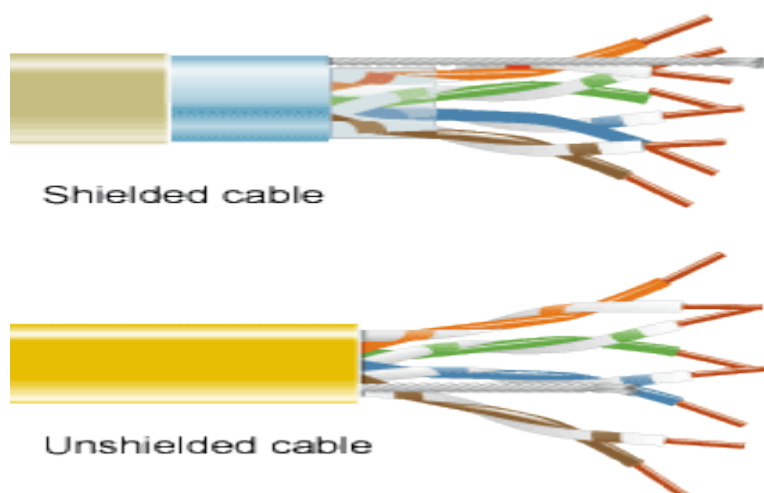
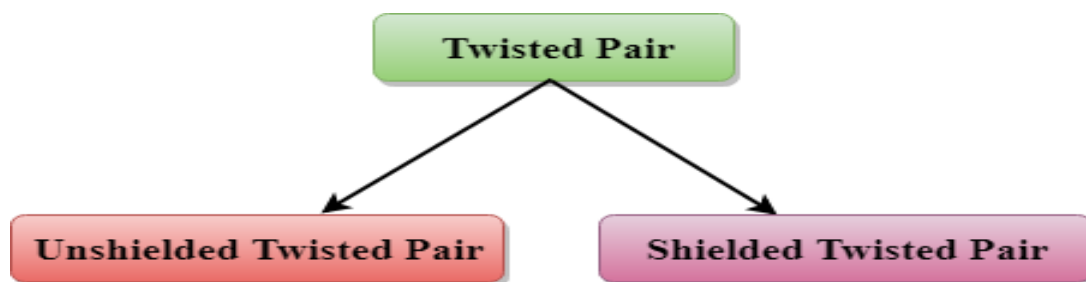
Guided Media– In guided media, transmitted data travels through cabling system that has a fixed path. For example, copper wires, fibre optic wires, etc.

Types Of Guided media:

The degree of reduction in noise interference is determined by the number of turns per foot. Increasing the number of turns per foot decreases noise interference.



Types of Twisted pair:



Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:

- **Category 1:** Category 1 is used for telephone lines that have low-speed data.
- **Category 2:** It can support upto 4Mbps.
- **Category 3:** It can support upto 16Mbps.
- **Category 4:** It can support upto 20Mbps. Therefore, it can be used for long-distance communication.
- **Category 5:** It can support upto 200Mbps.

Advantages Of Unshielded Twisted Pair:

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

Shielded Twisted Pair

A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics Of Shielded Twisted Pair:

- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

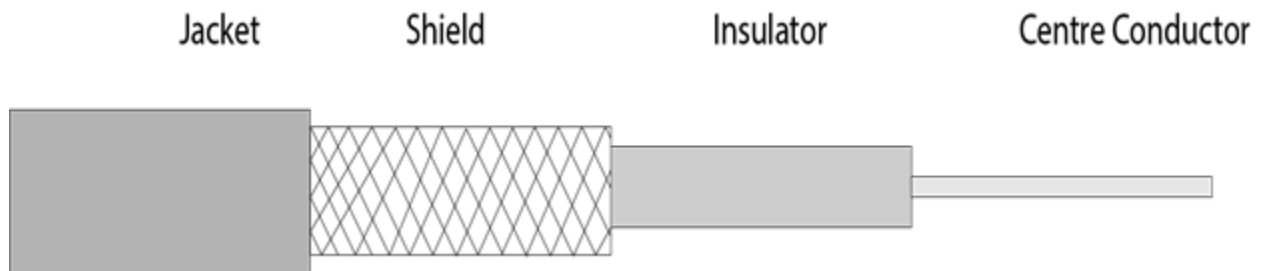
Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.

- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the **EMI**(Electromagnetic interference).



Coaxial cable is of two types:

1. **Baseband transmission:** It is defined as the process of transmitting a single signal at high speed.
2. **Broadband transmission:** It is defined as the process of transmitting multiple signals simultaneously.

Advantages Of Coaxial cable:

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

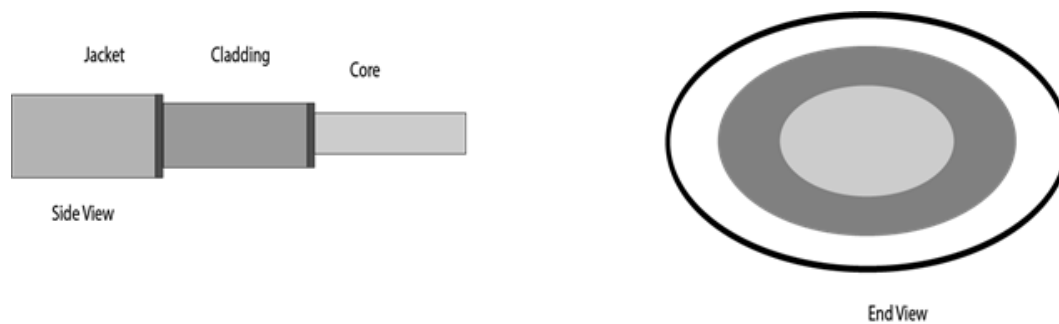
Disadvantages Of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

Fibre Optic

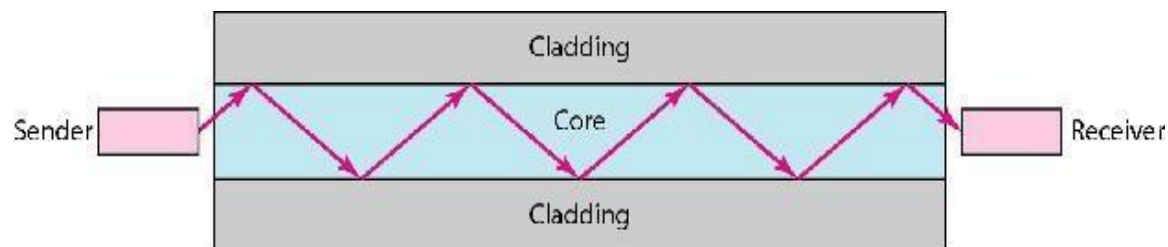
- Fibre optic cable is a cable that uses electrical signals for communication.
- Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:



Basic elements of Fibre optic cable:

- **Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.
- **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.
- **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.



Unguided Media – In unguided media, transmitted data travels through free space in form of electromagnetic signal. For example, radio waves, lasers, etc

Infrared

Low frequency infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc. Infrared signals can propagate within a room but cannot penetrate walls. However, due to such short range, it is considered to be one of the most secure transmission modes.

Microwaves

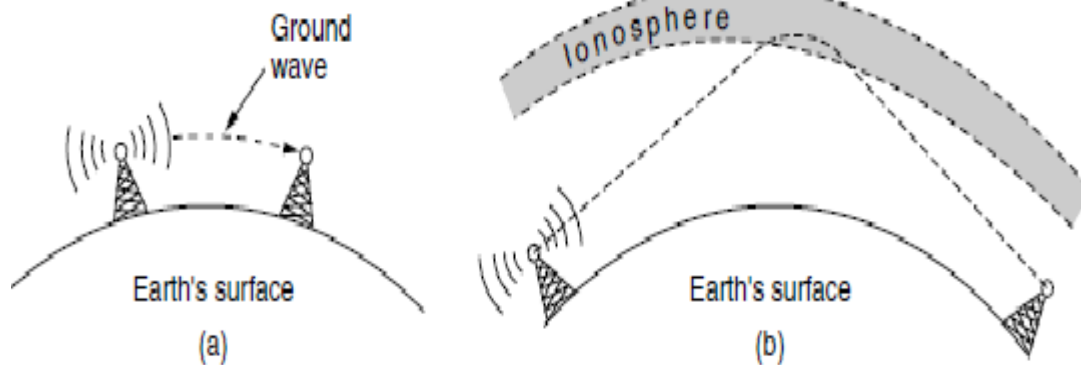
It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz. These are majorly used for mobile phone communication and television distribution

Radio Wave

Transmission of data using radio frequencies is called **radio-wave transmission**. We all are familiar with radio channels that broadcast entertainment programs. Radio stations transmit radio waves using **transmitters**, which are received by the receiver installed in our devices.

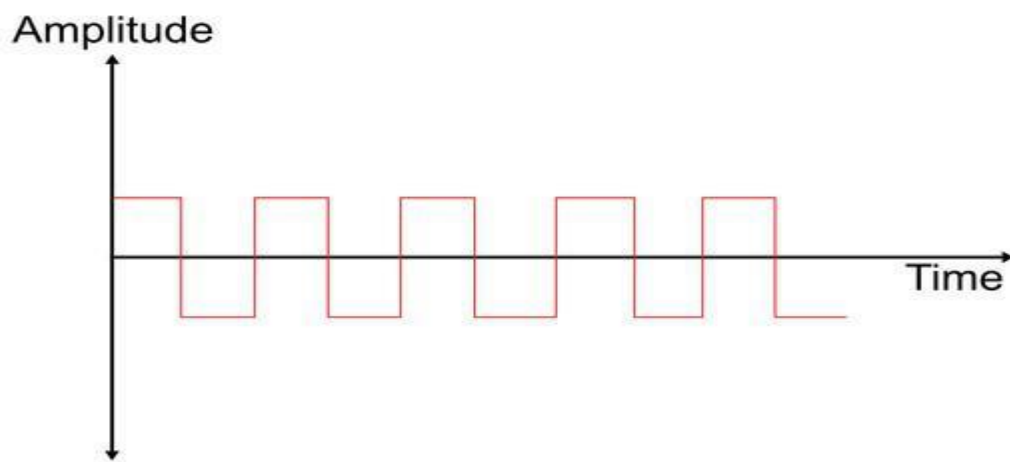
Both transmitters and receivers use antennas to radiate or capture radio signals. These radio frequencies can also be used for **direct voice communication** within the **allocated range**. This range is usually 10 miles.

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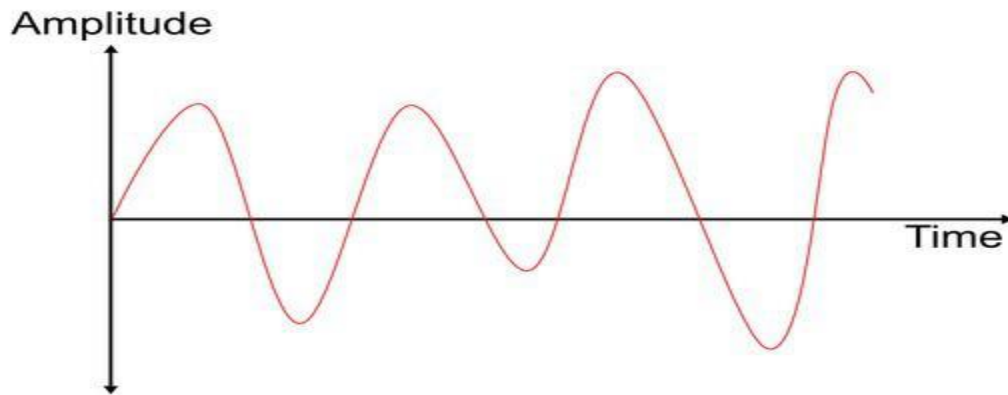


Analog to Digital conversion

A digital system is one whose signal has a finite number of discrete values. So, the digital system works on digital signals and is limited to binary values 0 or 1. Digital systems are used to process information in digital form. The digital system has wide applications in digital instruments like calculators, computers, Telephones, etc.



Analog system is one that uses continuous time signal or analog signal which is a sinusoidal waveform. Analog system transmits the output in their raw form reducing the time of translation. The amplitude of the signal varies continuously with the time. Analog signals are used to represent sound, temperature, light intensity etc.



1. Digital-to-Digital Conversion:(Digital data into digital signal data is sent to printer).

This involves representing digital information (like the binary code used by computers) using different digital signals. It's essentially encoding the data. For example, converting a series of bits (0s and 1s) into a specific sequence of voltage pulses for transmission over a wire is a digital-to-digital conversion. This process is also known as line coding.

2. Analog-to-Digital Conversion (ADC):(Analog to digital signal human voice transmitted to digital channel).

This process converts continuous signals (like sound waves or voltage fluctuations) into a digital representation. An analog-to-digital converter (ADC) takes a continuous analog signal as input and produces a corresponding digital value (a number). This is essential for processing real-world data with digital systems. The process typically involves sampling the analog signal at specific intervals and then quantizing the sampled values to discrete levels.

3. Digital-to-Analog Conversion (DAC):(Digital signal is to be transmitted over Telephone line).

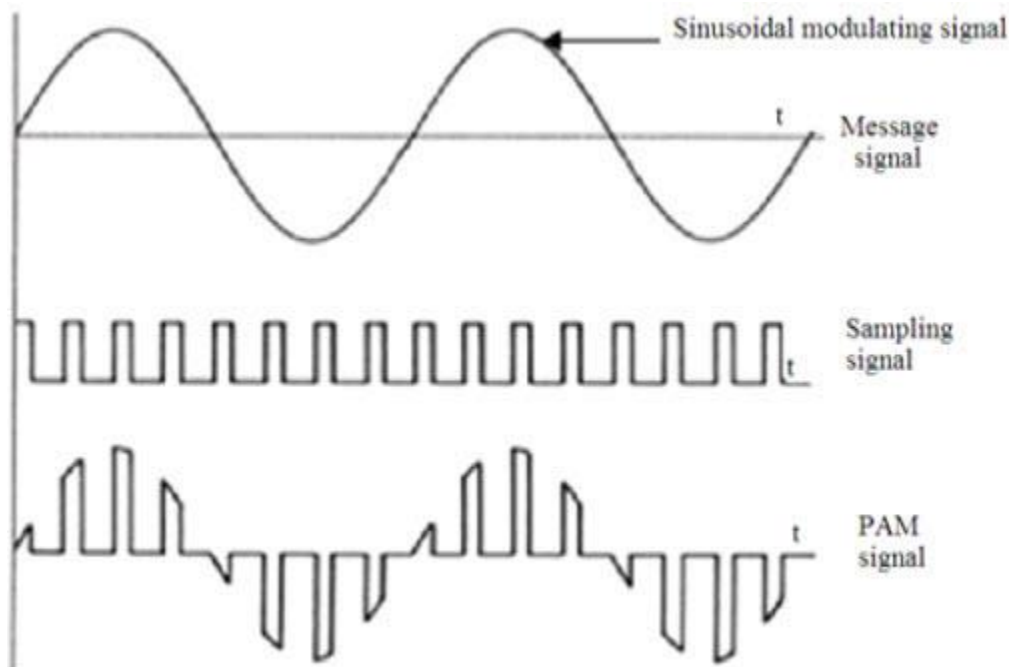
This converts digital data back into an analog signal. A digital-to-analog converter (DAC) takes digital data as input and produces a corresponding continuous analog signal. This is used when digital data needs to be converted back into a form that can be used by analog devices like speakers or motors.

4. Analog-to-Analog Conversion:Radio station convert voice signal (analog)into radio waves(analog).

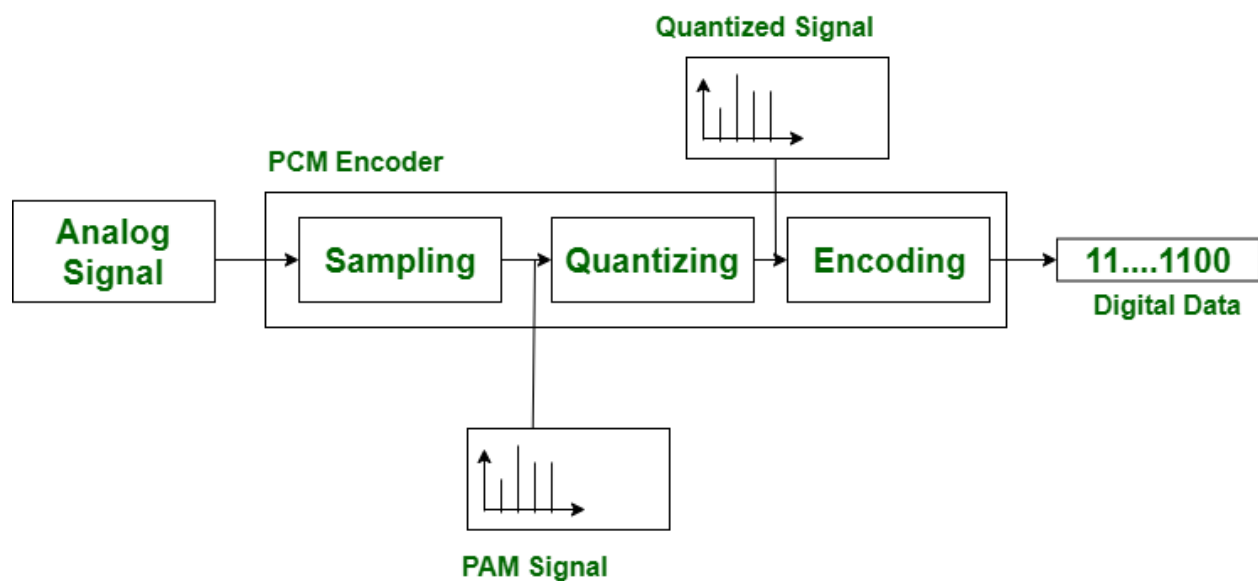
This involves modifying one analog signal into another analog signal. Examples include changing the amplitude, frequency, or phase of a signal. This is used in various applications like radio transmission, where different characteristics of a signal can be used to represent different data.

Pulse Amplitude Modulation (PAM) is a key modulation technique used in digital communication for transmitting analog data and is one of the most widely used types of analog-to-digital conversion. Its

process is simple where the amplitude of a sequence of pulses changes with the instantaneous amplitude of the analog message signal. The analog signal that is to be modulated is sampled by a sequence of pulses that are amplitude-modulated on the carrier to produce the amplitude-modulated pulses.



Pulse Code Modulation is the technique used for reworking analog signals into digital signals. PCM has a good or sensible signal-to-noise ratio. For transmission, Pulse Code Modulation wants high transmitter bandwidth. PCM technique is split into three elements, initial is the transmission at the provision end, second regeneration at the transmission path and conjointly the receiving end.



Block Diagram Of PCM

Delta modulation is an analog to digital and digital to analog signal conversion technique. [Delta modulation](#) is employed to realize high signal to noise ratio. It uses one bit [PCM](#) code to realize digital transmission of analog signal. With delta modulation, instead of transmit a coded illustration of a sample solely one bit is transmitted, that merely indicates whether or not the sample is larger or smaller than the previous sample.

it's the best type or simplest type of Differential Pulse Code Modulation. Delta modulation signal is smaller than Pulse Code Modulation system. If signal is large, the next bit in digital data is 1 otherwise 0.

